Compiler Construction Lab

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Lab 5

Task # 1

```
#include <iostream>

using namespace std;


struct Symbol

{

   string name;

   string type;

   string value;

   int size;

};


int main()

{

   Symbol table[10];

   int n;


   cout <<"--------Symbol table with memory size---------\n";

   cout << "How many variables do you want to enter? ";

   cin >> n;


   for(int i = 0; i < n; i++)
```

```
{
    cout << "\n--Entry " << (i+1) << "--\n";
    cout << "Enter data type (int/float/char): ";
    cin >> table[i].type;

    cout << "Enter variable name: ";
    cin >> table[i].name;

    cout << "Enter value: ";
    cin >> table[i].value;

    if(table[i].type == "int")
    {
        table[i].size = 4;
    }
    else if(table[i].type == "float")
    {
        table[i].size = 4;
    }
    else if(table[i].type == "char")
    {
        table[i].size = 1;
    }
    else
    {
        table[i].size = 0;
        cout << "Warning: Unknown data type, size set to 0\n";
```

```
        }
    }
    cout << "---------SYMBOL TABLE--------\n";
    cout << "Name\t\tType\t\tValue\t\tSize(bytes)\n";


    for(int i = 0; i < n; i++)
    {
        cout << table[i].name << "\t\t"
             << table[i].type << "\t\t"
             << table[i].value << "\t\t"
             << table[i].size << endl;
    }
    return 0;
}
```

Output:

```
C:\Users\Riphah\Desktop\Gull cc#5 1.exe
--------Symbol table with memory size---------
How many variables do you want to enter? 3

--Entry 1--
Enter data type (int/float/char): int
Enter variable name: price
Enter value: 20

--Entry 2--
Enter data type (int/float/char): char
Enter variable name: price
Enter value: gull

--Entry 3--
Enter data type (int/float/char): float
Enter variable name: price
Enter value: 3900
---------SYMBOL TABLE--------
Name            Type            Value           Size(bytes)
age             int             20              4
name            char            gull            1
price           float           3900            4

--------------------------------
Process exited after 39.54 seconds with return value 0
Press any key to continue . . .
```

Task # 2

```
#include <iostream>

using namespace std;



struct Symbol

{

   string name;

   string type;

   string value;

};



int main()

{

   Symbol table[10];
```

```
int n;

cout << "Symbol table with lookup\n";
cout << "How many variables do you want to enter? ";
cin >> n;

for(int i = 0; i < n; i++)
{
   cout << "\n--Entry " << (i+1) << "--\n";
   cout << "Enter data type: ";
   cin >> table[i].type;

   cout << "Enter variable name: ";
   cin >> table[i].name;

   cout << "Enter value: ";
   cin >> table[i].value;
}

cout << "------SYMBOL TABLE (CREATED)----------\n";
cout << "Name\t\tType\t\tValue\n";

for(int i = 0; i < n; i++)
{
   cout << table[i].name << "\t\t"
        << table[i].type << "\t\t"
        << table[i].value << endl;
```

```
}

char choice;
do
{
    string searchName;
    bool found = false;

    cout << "\n-------LOOKUP OPERATION-----\n";
    cout << "Enter identifier name to search: ";
    cin >> searchName;

    for(int i = 0; i < n; i++)
    {
        if(table[i].name == searchName)
        {
            cout << "\n✅ IDENTIFIER FOUND!\n";
            cout << "Name: " << table[i].name << endl;
            cout << "Type: " << table[i].type << endl;
            cout << "Value: " << table[i].value << endl;
            found = true;
            break;
        }
    }

    if(!found)
    {
```

```
        cout << "\nERROR: Identifier \"" << searchName << "\" not found in symbol table!\n";
    }

    cout << "\nDo you want to search another identifier? (y/n): ";
    cin >> choice;

  } while(choice == 'y' || choice == 'Y');

  cout << "\nThank you for using Symbol Table Lookup!\n";

  return 0;
}
```

Output:

C:\Users\Riphah\Desktop\Gull cc#5 2.exe

```
Symbol table with lookup
How many variables do you want to enter? 3

--Entry 1--
Enter data type: int
Enter variable name: price
Enter value: 23

--Entry 2--
Enter data type: string
Enter variable name: name
Enter value: gull

--Entry 3--
Enter data type: string
Enter variable name: gender
Enter value: female
------SYMBOL TABLE (CREATED)----------
Name            Type            Value
age             int             23
name            string          gull
gender          string          female

-------LOOKUP OPERATION-----
Enter identifier name to search: name

? IDENTIFIER FOUND!
Name: name
Type: string
Value: gull

Do you want to search another identifier? (y/n): y

-------LOOKUP OPERATION-----
Enter identifier name to search: gender

? IDENTIFIER FOUND!
Name: gender
Type: string
Value: female

Do you want to search another identifier? (y/n): y

-------LOOKUP OPERATION-----
Enter identifier name to search: int

ERROR: Identifier "int" not found in symbol table!

Do you want to search another identifier? (y/n): y

-------LOOKUP OPERATION-----
Enter identifier name to search: age

? IDENTIFIER FOUND!
Name: age
Type: int
Value: 23

Do you want to search another identifier? (y/n): n

Thank you for using Symbol Table Lookup!
```

Task # 3

```
#include <iostream>

using namespace std;


struct Symbol

{

   string name;

   string type;

   string value;

};


int main()

{

   Symbol table[10];

   int n, count = 0;


   cout << "Symbol table with Duplicate handling \n";

   cout << "How many variables do you want to enter? ";

   cin >> n;


   cout << "\n  Note: Duplicate variable names will be rejected!\n";


   for(int i = 0; i < n; i++)

   {

      string name, type, value;

      bool duplicate = false;
```

```
cout << "\n--Entry " << (i+1) << "--\n";
cout << "Enter data type: ";
cin >> type;

cout << "Enter variable name: ";
cin >> name;

for(int j = 0; j < count; j++)
{
    if(table[j].name == name)
    {
        cout << "ERROR: Duplicate identifier '" << name << "' already exists!\n";
        cout << " This entry will be rejected.\n";
        duplicate = true;
        break;
    }
}

if(!duplicate)
{
    cout << "Enter value: ";
    cin >> value;

    table[count].name = name;
    table[count].type = type;
    table[count].value = value;
    count++;
```

```
        cout << "Identifier inserted successfully.\n";
    }
}



cout << "------Symbol Table------\n";


if(count == 0)
{
    cout << "No valid entries in symbol table.\n";
}
else
{
    cout << "Sr.No.\tName\t\tType\t\tValue\n";


    for(int i = 0; i < count; i++)
    {
        cout << (i+1) << "\t"
            << table[i].name << "\t\t"
            << table[i].type << "\t\t"
            << table[i].value << endl;
    }
    cout << "Total unique identifiers: " << count << endl;
}


return 0;
}
```

Output:

```
C:\Users\Riphah\Desktop\Gull cc#5 3.exe
Symbol table with Duplicate handling
How many variables do you want to enter? 4

  Note: Duplicate variable names will be rejected!

--Entry 1--
Enter data type: int
Enter variable name: count
Enter value: 346
Identifier inserted successfully.

--Entry 2--
Enter data type: float
Enter variable name: temperature
Enter value: 53.5
Identifier inserted successfully.

--Entry 3--
Enter data type: string
Enter variable name: username
Enter value: AdminUser1
Identifier inserted successfully.

--Entry 4--
Enter data type: int
Enter variable name: count
ERROR: Duplicate identifier 'count' already exists!
 This entry will be rejected.
------Symbol Table------
Sr.No.  Name            Type            Value
1       count           int             346
2       temperature     float           53.5
3       username        string          AdminUser1
Total unique identifiers: 3

--------------------------------
Process exited after 58.99 seconds with return value 0
Press any key to continue . . _
```